

Problem 6.3. Farmer Shaun is producing sweet potatoes. He intends to harvest 10,000—cartons' worth in six months. His total costs are \$12.00 per carton.

He wishes to hedge using European put options. There are two puts on sweet potatoes with the exercise date in six months available: one with the strike price of \$13 per carton and another with the strike price of \$15 per carton. Their premiums are \$0.15 and \$0.18, respectively.

Assume that the prevailing risk-free interest rate is 4% effective for the half-year period.

At harvest time, in six months, it turns out that the sweet-potato spot price equals \$14. What would Farmer Shaun's profit be if he had decided to hedge using the \$13-strike put versus his profit if he had decided to use the \$15-strike put to hedge?

focus on the payoff w/out production costs.

unhedged : $S(T)$

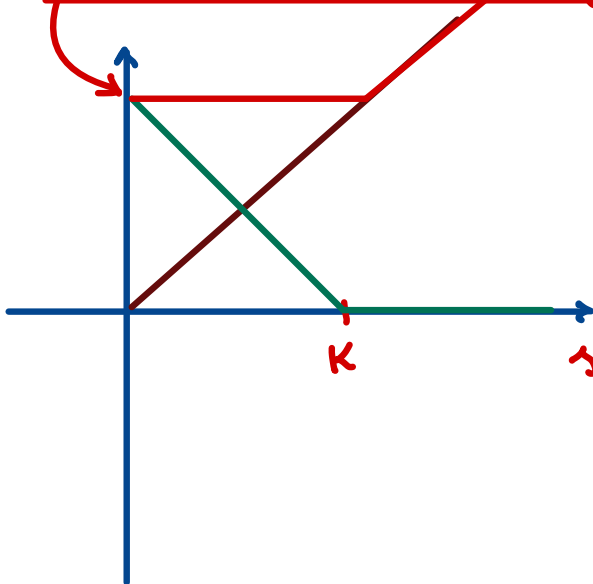
hedge : $(K - S(T))_+$

total hedged :

$$S(T) + (K - S(T))_+ = \begin{cases} K & \text{if } K > S(T) \\ S(T) & \text{if } K \leq S(T) \end{cases}$$

$$= \max(S(T), K)$$

FLOOR = long underlying + long put



The total profit:

$$\begin{aligned} & 10,000 \cdot \max(S(T), K) - 10,000 \cdot 12 - 10,000 \cdot V_p(0, K) \cdot (1.04) = \\ & = 10,000 (\max(S(T), K) - 12 - V_p(0, K) \cdot (1.04)) \end{aligned}$$

For $K=13$: _____

For $K=15$: